

Institute of Actuaries of India

Subject CA1-II – Actuarial Risk Management

April 2016 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- Under flexible benefit systems employees are offered the option to revise the level of an existing benefit or
- to choose different benefits which the employee “buys” either by reducing their pay or by giving up part of their existing benefits.
- Flexible benefit systems are useful when it is inappropriate to provide all employees in an organisation with the same benefit package.
- For example, a holiday package, lumpsum at retirement, medical benefits, etc.

[2]

ii)

Having a flexible benefit from employers perspective:

+ Employees can choose the benefits they need and it helps employers to attract and retain employees.

+As it provides benefits which employees’ value it increases their appreciation of benefits.

+Benefits offered in a cost effective manner as benefits which employees do not require is not offered.

+Annual enrolment to scheme means that employees are reminded of the benefits they receive and they appreciate the benefits

+Benefits are offered at no or very little cost.

-Some benefits take up rate may be so less that it could end expensive to administer.

-employees may choose benefits which they think they want but not what they actually need.

- Example expensive holiday over health care benefits.

- if third party chosen to administer a benefit mismanages it reflects badly on the employer.

- if the value of some option offered is costly for employees it may not appreciated by employees for eg.guaranteed benefit on retirement etc., .

[4]

[6 Marks]**Solution 2:**

- In this the unit growth rate is the most important assumption that will determine the guarantee.
- If the unit fund maturity is less than the GMMB then the guarantee will bite.
- Usually Deterministic model is a projection of one single outcome. But
- A range of deterministic scenarios can be projected ranging from optimistic to pessimistic scenarios.
- Probabilities for each outcome should be assigned but each scenario and probabilities will be subjective.
- In deterministic model it is also possible the most detrimental scenario may not be tested.
- Because of the uncertainty, stochastic model should be used which is useful for determining the likelihood of the guarantee applying and the cost of the guarantee.
- In the stochastic model, suitable parameters to project unit growth rate should be chosen.
- Co-relation can also be built between all economic parameters
- The model would be run using simulations.
- This could produce several thousand possible values for the unit fund at maturity, and hence the cost of the guarantee.

- For reserving purpose an appropriate probability of ruin needs to be chosen for example 1 in 100 in which case the reserve will be 99th percentile.

[6 Marks]

Solution 3:

i) The main benefits are:

- I. Reduction in claims volatility
 - i. ... leading to smoother profits
 - ii. Better capital management
 - iii. Efficient use of capital to write more new business
- II. Limiting large losses arising from
 - i. Single large risk
 - ii. One event
 - iii. Accumulation of events
 - iv. Concentration of risk (many risk at one geography or one kind of risks)
- III. Better manage solvency
- IV. Increased capacity to write larger risks and more new business
- V. Access to expertise of the reinsurer

[4]

ii)

a) Percentage risk retained:

Property1 = $40,00,00,000 / 100,00,00,000 = 40\%$ Property2 = $5,00,000 / 15,00,000 = 33.33\%$

[1]

b) Claims

Property1 = $50,00,00,000 * 60\% = 30,00,00,000$ Property 2 = $17,00,000 * 66.67\% = \text{Rs } 1133390 \text{ and Rs } 66,670$

In case of property 2, the claim amount of Rs 17,00,000 is higher than EML. The claim will still be split on the agreed proportion. This is a risk which is accepted by both parties. If the direct writer's underwriting continues to underestimate EML for property risks the reinsurer may likely refuse to do business in future.

[4]

[9 Marks]

Solution 4:

i)

- mortality, morbidity (if applicable)
- withdrawal / lapses
- investment income and gains
- expenses
- commission
- inflation
- taxation
- new business levels

- claim distribution- for eg large claim vs large no of small claims
- reinsurance
- change in valuation basis

[4]

ii)

- Any form of surplus distribution will depend on the communications made to the policyholders, smoothing policy and past practices relating to treatment of one-off surplus
- Smoothing policy of the company smoothens payment to different generations of policyholders and smooth volatile investment performance
- Surplus could be allocated to with-profit policyholders and shareholders or retained as working capital but in the end surplus distribution needs to ensure equity and ensure fairness between shareholders and policyholders.

One of surplus could be utilized as follows-

- This huge surplus due to sale of Investment property could be one time and not a regular surplus.
- Ideally one off surplus should be distributed as one-off bonus.
- Distributing the same as a part of normal RB declarations could increase expectations of policyholders which could be unsustainable in the long run.
- This could lead to customer and shareholder dissatisfaction, regulatory interference and loss of reputation.
- Defeats the very purpose of participating business which is delivering smooth bonuses.
- Alternatively as this surplus is one-off it could be paid as terminal bonus to avoid the above problems.
- Paying as terminal bonus will defer distribution of surplus and provide working capital for the company.

Surplus retained as working capital

- Deferring the distribution of surplus to policyholders reduces the level of guaranteed policyholder benefits and hence the capital required.
- Hence Company's free assets will improve
- And it could be used to protect against risk events,
- ensure solvency,
- finance profitable business growth
- and allow appropriate investment freedom
- This could still cause some dis-satisfaction as policyholders might feel denied due to the deferral especially if company is not able to use the surplus appropriately and makes a series of bad decisions
- Declaring an immediate one off bonus to reflect the one off performance could ensure that policyholders feel vindicated for putting their faith in the company and increase perception of fairness
- One off bonuses leads to higher one off increase in shareholder earnings
- Ideally a one off declaration utilizing some of the surplus and retaining the balance to smooth payouts in future could be a balanced approach.

[8]

[12 Marks]

Solution 5:**i) Risks for the tour operator could be-****Arising out of age of the travelers**

- Senior citizen tour will have all people over 60 yrs of age
- Potentially they could fall sick, due to hectic travel, change in environment, change in food
- They are more prone to accidents
- And potentially the damage could be severe
- They may be on medication for various existing ailments
.....and hence need for access to emergency medical aid, medicines
-Diet specifications could be rigid due to medical reasons and hence need to arranged for
- They may be more demanding and inflexible and hence air tight schedules with limited maneuverability
- The tour activities need to be suitable for their age and should not aggravate their existing health conditions

Other general risks

- Tour cancelling in the last minute by the customers due to emergency reasons
- Not enough senior citizens take up of such tours, may make it unviable for tour operator
- Weather change – flights cancelled and trains cancelled
- Not able to do planned outdoor activities due to sudden weather change, a landslide or blocked roads
- Riots, Strikes or curfew at the tourist spot
- Loss of luggage while travelling (at airport, station, on the way)
- Any change in tariffs/ rates between the date of booking by the client and the actual date of travel. Difficult to ask the client any more money
- A bad one-off accident and bad publicity can put reputation at stake

[7]

ii) Factors that should be considered:**Product design****Customers needs and interests**

- What all cover should be provided- The product design should at the minimum provide following covers-
 - Third party Liability insurance to cover risks to travellers, other third party due to negligence on part of the operator;
 - Product (Services) insurance; clients are not happy with the services and are claiming compensation or refund of money;
 - Travel cover for the senior citizens including hospitalization cover while on tour
 - Business interruption cover- if tours are cancelled/postponed due to inclement weather, man made events like airline strikes etc.

- Insurance cover for property(for eg., buses, cars etc.) of the operator and personnel of the tour operator (accidents, health cover etc.)
- Travel interruption covers like flight cancellation, delay, loss of baggage, valuables, passport, foreign exchange, burglary,
- Benefit structure
 - Certain covers would need to be offered on a fixed benefit basis while others would be on indemnity basis
 - Even in case of indemnity it cannot be unlimited always and some upper benefit limit could be in place.
 - Deductibles, excess, limits and sub-limits for each benefit for each coverage would need to be decided
- Policy terms and conditions and Exclusions
 - Exclusions within each cover would need to be decided
.....For eg., exclusion to health insurance cover, liability cover, subrogation, etc.
 - Policy terms and conditions need to be clear so that there is no issues at time of claim settlement
 - Terms may need to be separate for domestic and international travel
- Other product features like-
 - Minimum and maximum age of customers to be offered health cover
 - Period of cover – minimum and maximum days of cover would need to be decided, when does cover start and end would need to be decided
- Type of cover and level of cover will also depend on geography – for eg., tours to US could have different cover requirement than tours to Asia , hence product needs to be flexible to change as per requirements
- There is a risk of aggregation of risk (due to an accident or food poisoning etc) hence the design needs to mention what type of catastrophic events will be covered or not covered.

Pricing:

Profitability of the contract while pricing it

... competition vs marketability , decision needs to be made

Competition – Price and terms offered by competition

... if possible, the product should include valuable features or options that are not present in competitors' product.

Marketability

... it is a customized product, will there be enough business to justify the development cost

... since it is a leading tour operator, there is scope for good business

... also there might be other tour operators for whom this product can be marketed

once you have leading player as your client, marketing may be easier

This product should be designed in such a way that the capital requirements are not too onerous

For example, the limits on benefits, exclusions should be well thought through

Cross subsidy

- with other standard travel plans, other lines of business

[7]

[14 Marks]

Solution 6:

i) The advantage and disadvantages are

- +firms will have to hold capital appropriate to its inherent risks
- + ... which will improve the confidence in the market
- +can be explained to the management and measures can be taken to address these risks.
- + will improve security for policyholders
- + in turn help regulator to achieve their aim of policyholder protection, maintain confidence and promote orderly and efficient market.
- +Factor based supervisory reserves may lead to inadequate or higher provisions which is avoided.
- + good risk management rewarded
- Economic capital model may be new for firms making transition difficult
- regulator and insurers both may lack expertise to develop and understand complex models
- Time consuming, costly and difficult to develop
- Difficult for regulator to administer and monitor
- Difficult to understand for stakeholders

[6]

ii)

Economic capital is the amount of capital that a provider determines is appropriate to hold given its assets, its liabilities and its business objectives.

It will be determined based upon the risk profile of the individual assets and liabilities in its portfolio, the correlation of the risks and the desired level of overall credit deterioration that the provider wishes to be able to withstand.

The major risks to the life company are credit, market, longevity expense, and operational risk.

The first stage in a risk-based capital assessment for a provider is to produce an economic balance sheet.

An economic balance sheet shows:

- the market values of a provider's assets (MVA)
- the market values of a provider's liabilities (MVL)
- the provider's available capital, which is defined as $MVA - MVL$.

The *available capital* is then compared with the economic *capital requirement* to assess the provider's solvency status.

Market values of assets are usually easily and instantly available from the financial markets. Immediate annuities are invested mainly in a mixture of government and corporate bonds and market value is readily available.

The determination of a market value for a provider's liabilities is not so easy and a high level of judgment is required to determine market-consistent liability values.

A stochastic model (where at least the key parameters are modeled stochastically) or a deterministic model with scenario testing will generally be used to determine the value of liabilities

One approach is to determine the expected value of the unpaid liabilities on a present value best estimate basis and to add a risk margin to achieve sufficiency of reserves at certain confidence interval (say 75%). The liability cash flows will include the annuities and expenses for the existing portfolio.

A suitable stochastic model needs to produce internally-consistent possible future scenarios, *eg* for market risk reflecting downturns in investment performance and inflation and their impacts on reinvestment rates and new business.

An advantage of such a stochastic model is that it can automatically allow for correlations between different risk scenarios.

The company would project its balance sheet for each of a large number of future scenarios, which are intended to represent all the risks the company faces.

A risk measure, *eg* Value at Risk or Tail Value at Risk, would be used to determine the economic capital requirement, *eg* sufficient capital to maintain solvency in 1 years' time in 99.5% of scenarios.

[10]

[16 Marks]

Solution 7:

i)

Main types of liability insurance are:

- Employer's liability
- Motor third party liability
- Public Liability
- Product Liability
- Professional indemnity

[3]

ii)

Product liability would basically arise from faulty design, faulty manufacture.

Information the company should obtain includes:

- Products being currently sold for g., diesel, petrol, battery operated, any new non fossil fuel car
- Design /technology being used-
.....tried and tested or any new design being used for first time
- Manufacturing process-
.....ISO certified, long experience, set process, systems and controls,
- Manufacturing- automated and manual,
- Quality control measures to test the products
- Materials being used- where is it procured, vendors, their reputation,
- All in house or some part of manufacture outsourced
- If outsourced, how QC is ensured, reputation of outsourced vendor
- Past history of product recalls and reasons for recall
- Any potential dangerous components within the product, and how quickly they deteriorate-
- The material used for packaging and transportation
- Where the product is sold, as some countries are more litigious than others

[5]

iii)

The company is offering the product for the first time and so will not have past data and expertise on which to base the pricing. Pricing needs to be prudent but...

Marketability

- Attractiveness to the company buying it, must not be too prudently priced that it is uncompetitive
- Competition pricing and rating factors for such products
 - Is there a similar product in the market, what are the competitors charging
 - Important that risk classification/rating factors used is adequate compared to competitors market as there is an anti-selection risk-the risk of applications for insurance who know the product represents a greater risk than has been reflected in the premium
- Any regulatory restrictions or tariffs to be guided by
- Past experience of the industry /company may need to be interpreted with care, as future claims experience would be impacted by-
 - an increasingly litigious society
 - a greater frequency of court awards in favour of the consumer
 - latent claims – ie claims that were not anticipated when the insurance was sold and hence were not reflected in the premium

- It is likely that reinsurance will be used since this is a new product offering. Costs of reinsurance to be allowed for in pricing.
- What other loadings to add
 - What would be the expenses especially claims related in this product can bear
 - Expected Volume of business expected to be sold- to this manufacturer and any customer from motor industry-how can development costs be spread
 - Business mix of the manufacturer in terms make and model of vehicle, geography it will be sold(due to legal practices)

[9]

[17 Marks]**Solution 8:****i) Advantages of DC over DB****For the employer:**

- Cost of the liability is known; usually as a percentage of wage bill.
- DB scheme liability (depending on the final benefit promised) could be a unlimited liability
- In a DB scheme there may still be an uncertainty of future salary escalations and
- With changing economic and business environment; the cost of the liability can be very volatile
.... and might bite when the business is not doing good
- DC scheme has stable contribution rate compared to a DB scheme
- No need for yearly actuarial valuation of liabilities and determining contributions to fund
- It is easy to maintain, individual member-wise account.
- Also easy to make it a contributory scheme
.... employee's contribution goes into his account
- Instead in a DB scheme, there is an element of one generation paying for another
- Investment risk is not on the employer; bad investment performance will affect the employee account growth at the same time, a good investment performance will improve the fund growth in the employee account
- Easy to administer

For the employee:

- If the employer allows it to be contributory, then employees can also save for their retirement
- It allows flexibility for varying contributions (one can contribution varying proportion of salary based on changing liability profile of the individual in one's lifetime)
- They have more control on the investments, in the form of choice of funds
- Any investment upside directly gets credited into their retirement account
- It feels more secure to see your account progress to retirement; than depend on a defined benefit at retirement
- It is difficult for the employees to judge the funding levels of a DB scheme.

- What if all the funds are used to pay for the current pensioners and not much is left when young employees are going to retire in future
- Portability of retirement account becomes easy in a DC scheme than a DB one

[8]

ii)

If the benefits are defined and guaranteed as far as the past service is concerned, the ultimate liability will be with the employer.

The value of the liability will only give the employer an estimate about the total cost

The amount they need to set aside

The final cost will depend on how the future assumptions unwind.

Also bear in mind any legislation or scheme rules that should be taken account of in these calculations

However, in order to estimate the DB liability, the factors to keep in mind are:

Benefits to be valued

- Need to consider all the benefits promised under the scheme
 - .. including various options and guarantees given
 - ...value the costliest option
- Benefits will vary for class of members
 - Active members
 - Deferred members
 - Current pensioners
- Future enhancements promised
 - ... for existing pensioners what should be the assumption for future pension increases... for inflation protection (if promised; inflation linked)
 - ... any PRE set in that respect
- Other benefits on death; should they continue; will spouse pension on death etc be payable
- Is there any part of the benefit governed by any statutory benefit (so some minimum rules should be checked)

Existing Fund/ Assets

- Where are the current funds invested
 - ... how to value them
 - ... do they have market value
 - ... any insurance policy taken
- Is it fully funded; Are the employers in a condition to fully fund the past liability; any cash crunch
- If the fund is in surplus, can it be refunded back to the employer or
- Should be used to enhance benefits
- What does the statute say or the scheme rule say

- Should the investment manager change the asset profile; given that there won't be any further contributions into this fund.
- The earlier asset allocation might be with the idea that it is an open continuing fund

Data for calculation

- Need good quality data of all employees, salaries, DOB, etc
- You may get full employee wise data or in some cases only summary level information
- No of members in each class (active, deferred and pensioners)
- Scales of benefit for various classes once liability is frozen

Assumptions

Future salary escalations ; should it be taken for benefit calculations

- What does revised scheme rules state, any regulatory requirement that future salary escalation has to be allowed, if yes, how much
- Pensions in payments-
- If pensions enhancements were given in past what does revised scheme rules state, any regulatory requirement that future pension enhancements has to be allowed, if yes, how much

Is Spouses pension to be allowed, if yes need to make suitable assumptions for spouse pension benefits and pension enhancements

Mortality, attrition rate and retirement rate

- Should these assumptions vary by class of employees (like different attrition rates for junior staff and senior management staff)

Discount rate used in calculations

- Should it be based on current asset holding or long term interest scenario
- Need to bear in mind, this assumption can be a vital in the discounted value of the liability

Should be maintain some margin in these assumptions

Expenses

- Costs of managing the scheme
- Future inflationary impact on expenses

[12]

[20 Marks]
